

The second row

Same transverse structure, new colour — does that change the answer?

$$K^+ \equiv p^+ + k^+, \quad \xi \equiv p^+/k^+$$

0. Verdict

Both quoted equations are correct, by exactly the argument verified for the previous row — nothing in the derivation changes, because the ket bracket differs only in its *colour* structure.

The transverse situation is also unchanged: no ultraviolet divergence, but the $\frac{1}{p^+}$ term's z -integral is again the BK real-emission kernel, falling as $1/z^2$. So the $+$ prescription is again an exact identity you may write, and again it does not produce two separately finite pieces.

The new colour structure does not rescue it (§3) — though it does buy you something else, at $w \rightarrow y$ (§4).

1. Why the derivation is unchanged

Compare the two ket brackets:

$$\text{previous: } U^{bc}(z) \bar{\mathfrak{A}}_i^a(k^+, -w) \bar{\mathfrak{A}}_j^c(p^+, -z) - \bar{\mathfrak{A}}_i^a(k^+, -w) \mathfrak{A}_j^b(p^+, -z)$$

$$\text{this row: } U^{bc}(z) \bar{\mathfrak{A}}_j^c(p^+, -z) \bar{\mathfrak{A}}_i^a(k^+, -w) - U^{ac}(w) \bar{\mathfrak{A}}_j^d(p^+, -z) U^{db}(z) \mathfrak{A}_i^c(k^+, -w)$$

Each term still contains exactly one $\mathfrak{A}(p^+, -z)$ and one $\mathfrak{A}(k^+, -w)$, at the same momenta and the same positions. The bra bracket and $\bar{\mathfrak{C}}^\dagger$ are untouched. So the entire assembly — three $\mathfrak{A}$'s and one $\mathfrak{C}$, giving g^4 and $-i$; the numeric factor $\frac{1}{4\pi^4}$; the cancellation of $\sqrt{p^+k^+}$ leaving $\frac{1}{(p^++k^+)^2}$; the delta Jacobian $(\frac{K^+}{k^+})^2$ turning that into $\frac{1}{k^{+2}}$; and the index bracket $[\delta_{k'm}\delta_{ij} - \frac{K^+}{p^+}\delta_{jm}\delta_{ik'} - \frac{K^+}{k^+}\delta_{im}\delta_{jk'}]$ — carries over verbatim. Likewise the w' integration.

Two typos in the measures. Both displayed equations integrate over $\int_{w,w',y',z,y,x}$ and $\int_{w,y',z,y,x}$, but x' appears in the integrand, in $\frac{(x'-y')^{k'}}{(x'-y')^2}$ and $\rho^{e'}(x')$. It should be $\int_{w,w',y',z,y,x,x'}$ and $\int_{w,y',z,y,x,x'}$. (The same slip is in the previous row's first equation.)

2. Transverse divergences: identical to before

The four kernels are the same four kernels,

$$\frac{(y'-z)^m}{(y'-z)^2} \frac{(x'-y')^{k'}}{(x'-y')^2} \frac{(y-w)^i}{(y-w)^2} \frac{(x-z)^j}{(x-z)^2},$$

so every statement about the transverse integrations carries over:

- **No ultraviolet divergence.** Each point is a coincidence of exactly one kernel, so the integrand is $O(1/\rho)$ there, and the leading piece integrates to zero by parity. The collapses of three and four points converge too, by measure counting.
- **The $\frac{1}{p^+}$ structure is the problem.** $\delta_{jm}\delta_{ik'}$ contracts the kernels into $\frac{(y'-z)\cdot(x-z)}{(y'-z)^2(x-z)^2}$ — the BK real-emission kernel, falling only as $1/z^2$. For $p^+ \neq 0$ the phase $e^{-i\xi k\cdot(y'-z)}$ cuts it at $|z| \sim k^+/(p^+|k|)$; at $p^+ = 0$, where the subtraction sits, nothing does.

3. Does the new colour structure save the large- $|z|$ region?

This is the one thing genuinely worth re-checking, since a colour factor that vanished as $|z| \rightarrow \infty$ would kill the logarithm. It does not. Taking $U(z) \rightarrow 1$ (adjoint identity) and evaluating with random adjoint Wilson lines:

bracket	limit at $U(z) \rightarrow 1$	magnitude
bra (unchanged)	$\delta^{bd} [U^{e'c'}(y') - U^{e'c'}(x')] \rho^{e'}(x')$	1.9703
ket, previous row	$U^{ad}(y) \rho^d(y) [U^{be}(x) \rho^e(x) - \rho^b(x)]$	7.3631
ket, this row	$U^{be}(x) \rho^e(x) [U^{ad}(y) - U^{ad}(w)] \rho^d(y)$	4.7409

each verified to 2.7×10^{-15} . **All three are z -independent and nonzero**, so the coefficient of the large- $|z|$ logarithm survives in both rows. The divergence is exactly as before, and so is the remedy: the virtual diagrams, which replace $-2\mathcal{K}$ by $\mathcal{M} = \frac{(x-y')^2}{(x-z)^2(y'-z)^2}$ and ρ^{-2} by ρ^{-4} .

4. What the new colour structure does buy

Notice what the limit is: a **difference of Wilson lines at y and w** . The previous row's ket bracket gave a difference in x instead. This one therefore vanishes as $U(w) \rightarrow U(y)$ — linearly, as checked with $U(w) = U(y)e^{\epsilon X}$, X antisymmetric:

$ U(w) - U(y) _{\text{max}} = 3.375\text{e-}01$	$\text{max} \text{bracket} = 5.5751\text{e-}01$
$ U(w) - U(y) _{\text{max}} = 2.870\text{e-}02$	$\text{max} \text{bracket} = 5.5111\text{e-}02$
$ U(w) - U(y) _{\text{max}} = 2.803\text{e-}03$	$\text{max} \text{bracket} = 5.5770\text{e-}03$
$ U(w) - U(y) _{\text{max}} = 2.797\text{e-}04$	$\text{max} \text{bracket} = 5.5835\text{e-}04$

That is a real-virtual structure in the (y, w) pair, and it softens precisely the coincidence where the kernel $\frac{(y-w)^i}{(y-w)^2}$ is singular. It is the analogue, in the w integration, of what the virtual diagrams must still do for the z integration. Useful — but it says nothing about z .

5. So, on the + prescription

Yes, you can write the same + prescription, in exactly the same form:

$$\frac{1}{p^+} \longrightarrow \left[\frac{1}{p^+} \right]_+ + \delta\text{-part} \times \log \frac{\vee - k^+}{\Lambda},$$

with the second term carrying the $\delta_{jm}\delta_{ik'}$ contraction $(y'-z) \cdot (x-z) (x'-y') \cdot (y-w)$, no ξ -phase, prefactor $\frac{1}{8\pi^5}$ and a + sign, and **this row's** colour brackets. Keeping the lower limit at Λ makes it exact.

And the same caveat, unchanged. Neither resulting term is finite on its own: both diverge logarithmically at large $|y' - z|$, with equal and opposite coefficients. So write it if you want the rapidity logarithm displayed, but call the first term the soft-subtracted remainder, not a finite contribution — and do not absorb the second into the evolution until the virtual diagrams are in.

zint/src/colour_limits.py — the colour limits, written to zint/RESULTS_colour.txt; zint/src/assemble_row.py — the assembly, which is unchanged.